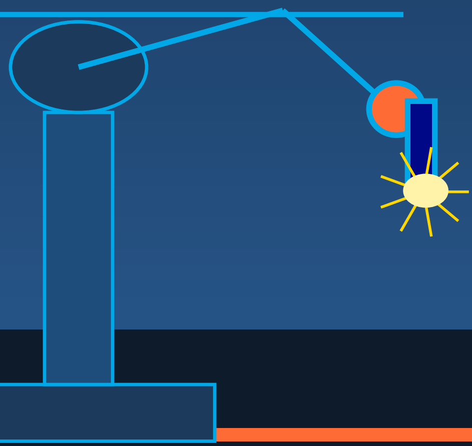
WELDING ROBOT

TECHNICAL DOCUMENTATION

Engineering Design • Control Systems • Manufacturing



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00 Overview & Specifications

System summary and key parameters

System Overview

The WR-6A Welding Robot is a 6-degree-of-freedom industrial articulated arm designed for high-precision MIG, TIG and plasma arc welding operations. It combines advanced servo-driven joints, real-time EtherCAT communication, integrated sensor fusion, and model-based trajectory planning to achieve sub-millimetre repeatability across complex 3-D weld paths.

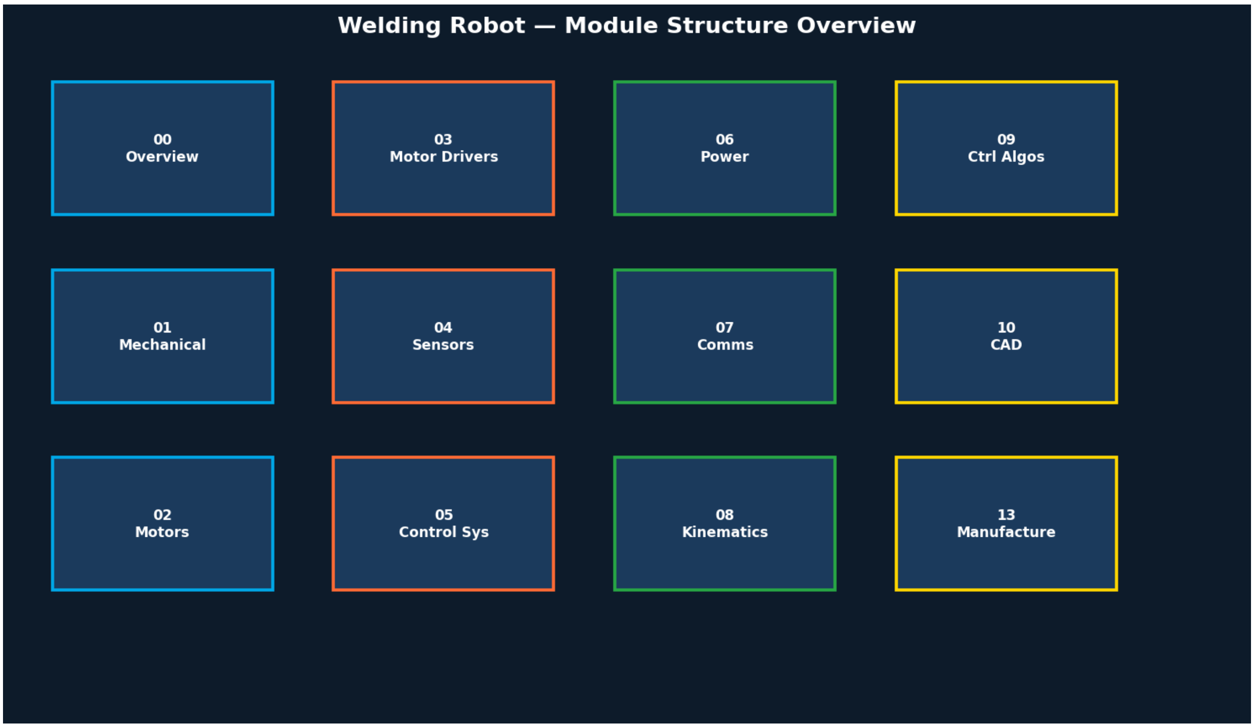

Figure 1 – Module Structure Overview

Key Specifications	
Degrees of Freedom	6 (rotary joints J1–J6)
Payload Capacity	8 kg at full extension
Max Reach	1 400 mm
Repeatability	± 0.05 mm
Weld Processes	MIG / MAG, TIG, Plasma Arc
Cycle Time (typ.)	< 3 s positioning, continuous weld
IP Rating	IP54 (body), IP67 (wrist & TCP)
Power Supply	3-phase 400 V AC, 50/60 Hz, 4.2 kW
Operating Temperature	0 °C to 50 °C

Communication	EtherCAT, Modbus TCP, ProfiSafe

01

Mechanical System

Structural design, materials and joint configuration

Structural Architecture

The robot frame is constructed from high-grade aluminium alloy (7075-T6) for the upper arm and forearm links, and ductile cast iron for the base and shoulder housing. Harmonic drive gearboxes are used at J1–J3 for high torque density; joints J4–J6 use lightweight crossed-roller bearing arrangements.

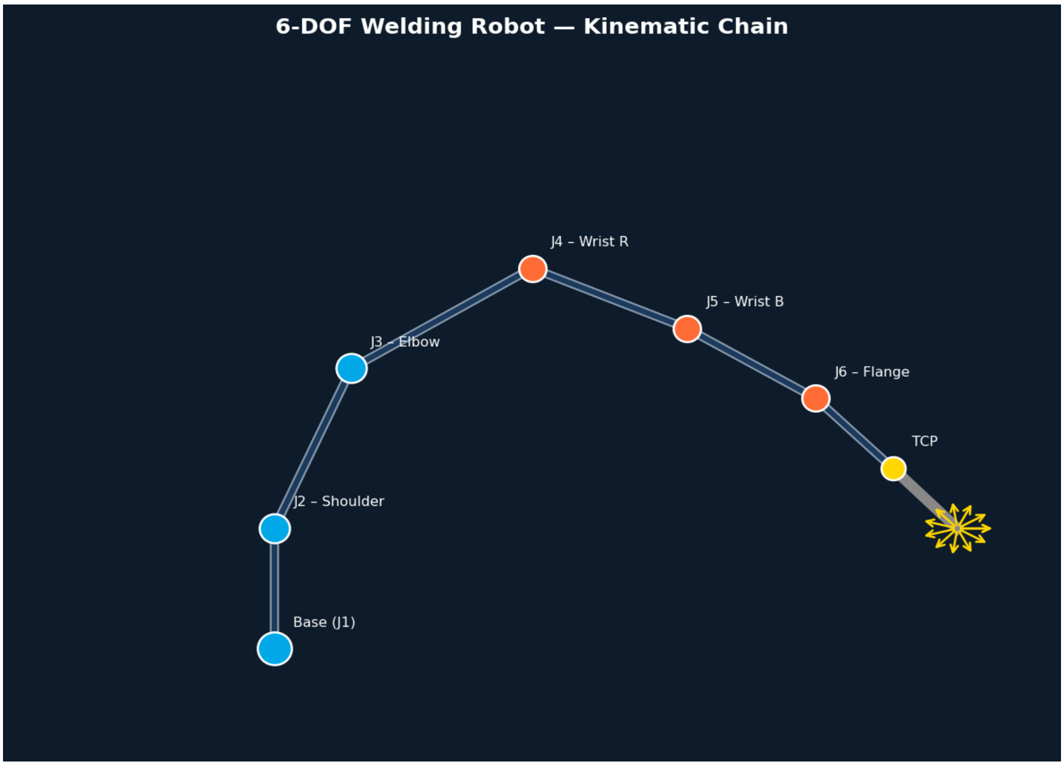

Figure 2 – 6-DOF Kinematic Chain and Joint Labels

Mechanical Specifications	
Base Mounting	ISO 9283 — ø 200 mm bolt circle, M16×8
Arm Material	7075-T6 Aluminium / Ductile Cast Iron
Gearbox Type J1-J3	Harmonic Drive® CSF-32 (ratio 1:160)
Gearbox Type J4-J6	Planetary — Neugart PLE 060 (1:50)
TCP Interface	ISO 9409-1 — ø 50 / M5×4 pattern
Robot Weight	127 kg (without TCP tooling)
Wrist Singularity	Software-avoided via path blending

02

Motors

Servo motor selection and performance characteristics

Motor Configuration

All six joints employ brushless DC servo motors with integrated 23-bit absolute encoders. J1 and J2 use high-torque frameless kit motors; J3–J6 use compact housed servo units. Each motor is matched to the harmonic or planetary gearbox stage for optimal bandwidth and back-drivability.

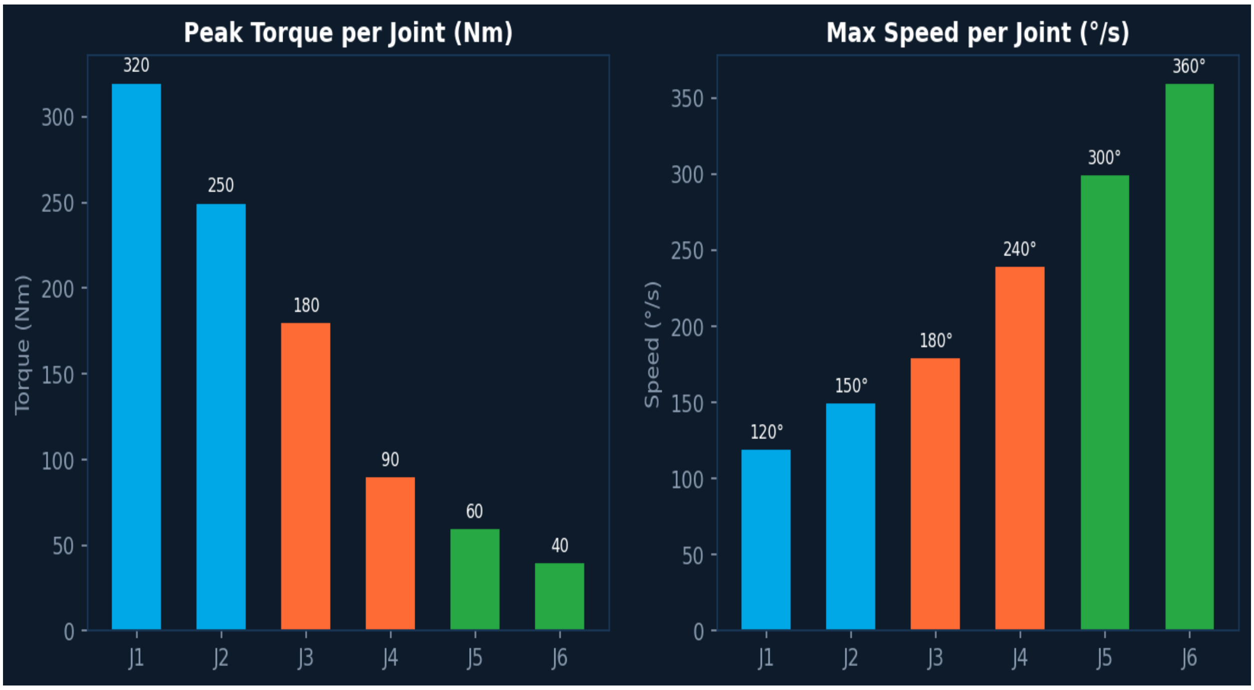

Figure 3 – Peak Torque and Max Speed per Joint

Joint	Motor Model	Rated Power	Peak Torque	Encoder
J1	Kollmorgen AKM65	2.2 kW	320 Nm (at output)	23-bit Abs.
J2	Kollmorgen AKM55	1.5 kW	250 Nm	23-bit Abs.
J3	Kollmorgen AKM45	1.0 kW	180 Nm	23-bit Abs.
J4	Maxon EC-i 40	400 W	90 Nm	19-bit Abs.
J5	Maxon EC-i 40	250 W	60 Nm	19-bit Abs.
J6	Maxon EC-i 30	150 W	40 Nm	19-bit Abs.

03 Motor Drivers

Drive electronics and EtherCAT integration

Drive Architecture

Each axis uses a dedicated EtherCAT-capable servo drive. J1–J3 use Kollmorgen AKD2G dual-axis drives; J4–J6 use Maxon EPOS4 compact drives. All drives operate in CSP (Cyclic Synchronous Position) mode at a 1 ms fieldbus cycle time, providing deterministic torque, speed, and position loop closure.

Drive Parameters	
Fieldbus Protocol	EtherCAT (IEC 61158) — daisy-chain topology
Cycle Time	1 ms (position loop), 250 µs (current loop)
Control Mode	CSP / CSV / CST (CiA 402)
Bus Voltage	48 V DC (J4–J6), 340 V DC (J1–J3)
Peak Current J1	42 A
Peak Current J4–J6	15 A
Regen Resistor	100 W shared bus resistor, thermal-protected
Safety Function	STO (Safe Torque Off), SLS (Safely Limited Speed)

04 Sensors & Feedback

Perception, seam tracking and state estimation

Sensor Suite

The welding robot integrates five categories of sensors to achieve closed-loop weld quality control: joint encoders for proprioception, a 6-axis force/torque sensor at the wrist, a structured-light vision camera for seam detection, an arc voltage/current sensor for process feedback, and thermocouples for thermal monitoring.

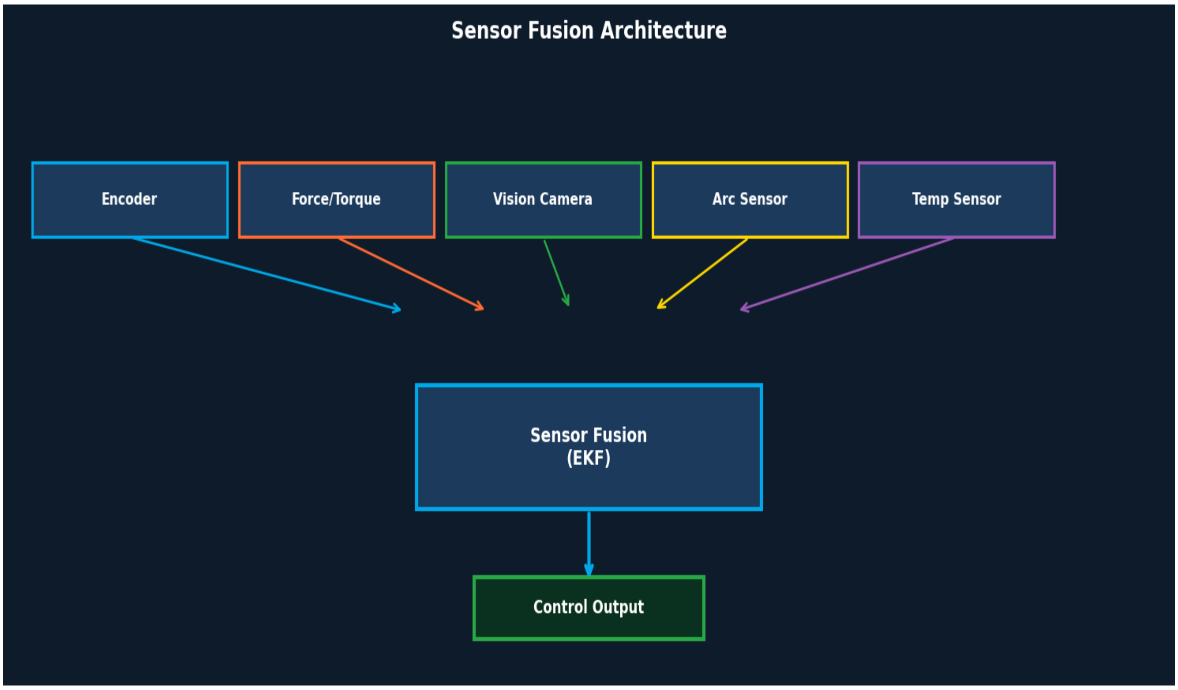

Figure 4 – Sensor Fusion Architecture

Sensor Specifications	
Joint Encoders	Renishaw RESOLUTE — 23-bit, ±0.1 arc-sec
Force/Torque	ATI Mini45 — 6-axis, 500 Hz sampling
Vision Camera	SICK Ranger3 — 3D laser profilometer, 2 kHz
Arc Sensor	Lincoln Electric iWave — V/I feedback, ±0.5%
Temperature	K-type thermocouple — TCP and weld pool monitoring
Fusion Algorithm	Extended Kalman Filter (EKF) — 500 Hz

05

Control System

Real-time controller hardware and software stack

Real-Time Control Platform

The robot controller runs on a dedicated industrial PC (Beckhoff C6030) with a real-time EtherCAT master (TwinCAT 3). The application layer implements a modular ROS 2 (Humble) stack with MoveIt 2 for motion planning, and a custom weld process server communicating via DDS. Safety functions are delegated to an independent Pilz PNOZmulti 2 safety PLC.

Control Stack Summary

IPC Hardware	Beckhoff C6030 — Intel Core i7-11th Gen, 32 GB RAM
RTOS	TwinCAT 3 / XAR (EL6695-1001 EtherCAT bridge)
Motion Framework	ROS 2 Humble + MoveIt 2
Weld Process Ctrl	Custom C++ node — arc length, wire feed, gas flow
Safety Controller	Pilz PNOZmulti 2 — PLe Cat.4 (ISO 13849)
HMI	Web-based React dashboard over REST/WebSocket
Data Logging	InfluxDB + Grafana — 1 kHz weld telemetry
Simulation	Gazebo Ignition + URDF model

06

Power System

Electrical supply, distribution and protection

Power Architecture

The robot cell operates from a 3-phase 400 V AC / 50 Hz supply. A main isolation transformer and EMC filter precede the distribution panel. Individual DC bus rails supply the servo drives, while a 24 V SMPS feeds control electronics and sensors. Regenerative braking energy from J1–J3 is shared on a common DC bus with a bleed resistor for overvoltage protection.

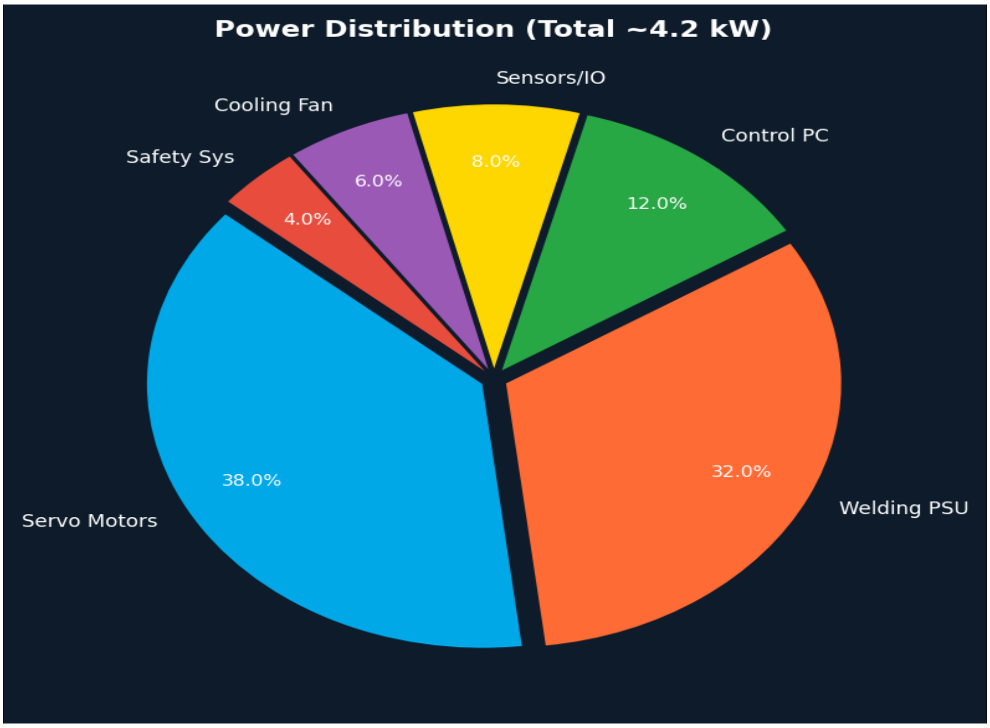

Figure 5 – Power Distribution by Subsystem

Power Specifications	
Mains Input	3-phase, 400 V AC ±10%, 50/60 Hz
Total Installed Load	4.2 kW (continuous), 9.5 kW (peak, 500 ms)
DC Bus — Drives	340 V DC (J1–J3), 48 V DC (J4–J6)
Control Supply	24 V DC, 10 A — Siemens SITOP PSU200M
Welding PSU	Fronius TPS 400i — 400 A, water-cooled
Protection	MCB, RCCB (30 mA), surge arrestors, STO relay
UPS	500 VA for IPC and safety PLC — 15 min backup

07 Communication Architecture

Fieldbus, networking and protocol stack

Network Topology

Real-time control uses EtherCAT in a daisy-chain topology connecting all six servo drives at a 1 ms cycle. The IPC also hosts a GigE Ethernet switch for non-RT traffic (vision, HMI, logging). Safety signals are exchanged with the PLC over ProfiSafe. External integration is via Modbus TCP and OPC-UA for MES/SCADA connectivity.

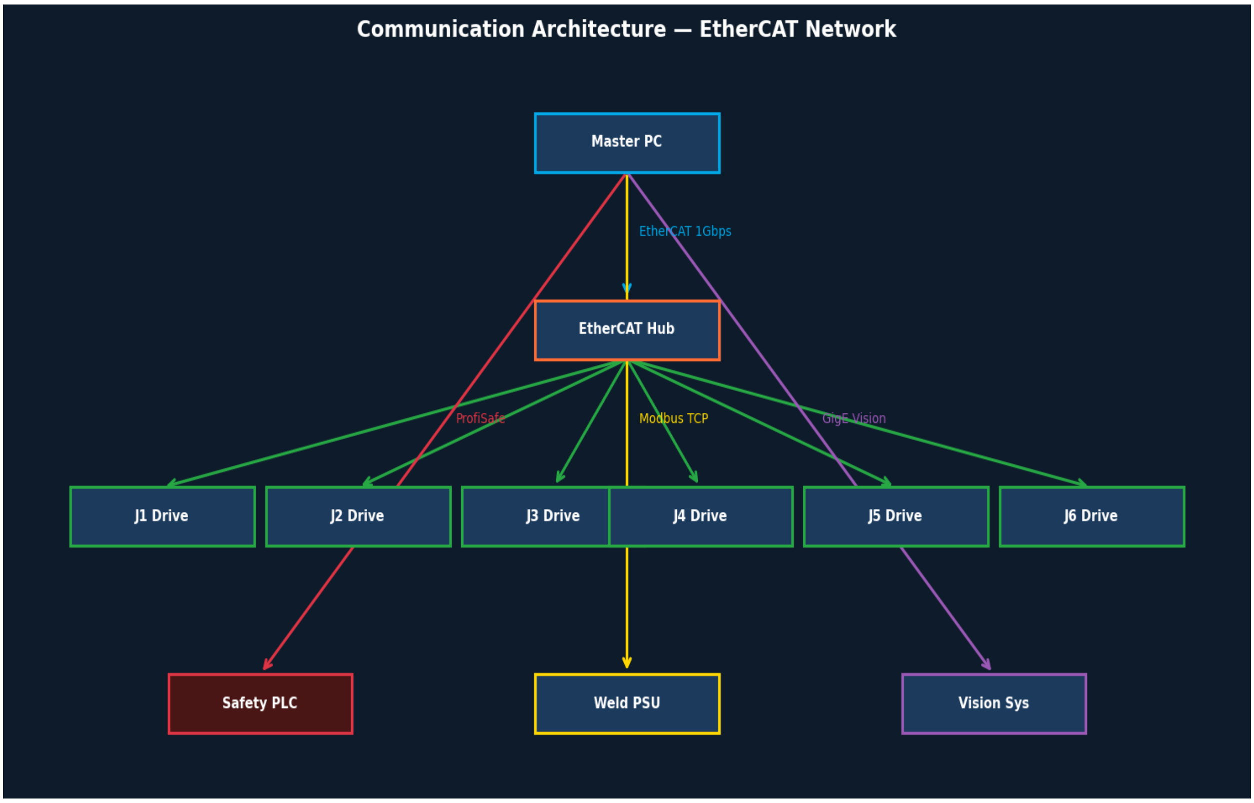

Figure 6 – Communication Architecture Diagram

Protocol Reference

EtherCAT	IEC 61158 — 1 Gbps, 1 ms cycle, daisy-chain
ProfiSafe	IEC 61784-3-3 — PLC ↔ IPC safety channel
Modbus TCP	RFC 793 — weld PSU and auxiliary I/O
GigE Vision	AIA standard — 3D camera data, 1 Gbps
OPC-UA	IEC 62541 — MES integration, 100 ms poll
DDS / ROS 2	OMG DDS — inter-node ROS 2 middleware
WebSocket	RFC 6455 — HMI real-time dashboard

08

Kinematics

Forward / inverse kinematics and workspace analysis

Denavit-Hartenberg Parameters

The WR-6A kinematic model uses the modified DH convention (Craig 1986). Closed-form analytical IK is solved using the Pieper condition (wrist intersection at a point), enabling real-time IK at > 5 kHz.

Joint	a (mm)	α (°)	d (mm)	θ (°)	Range (°)
J1 (Base Rotation)	0	0	400	θ1	±180
J2 (Shoulder)	25	-90	0	θ2	-95 / +130
J3 (Elbow)	455	0	0	θ3	-180 / +75
J4 (Wrist Roll)	-35	-90	420	θ4	±270
J5 (Wrist Bend)	0	90	0	θ5	-130 / +130
J6 (Flange Rotation)	0	-90	80	θ6	±360

Workspace Envelope

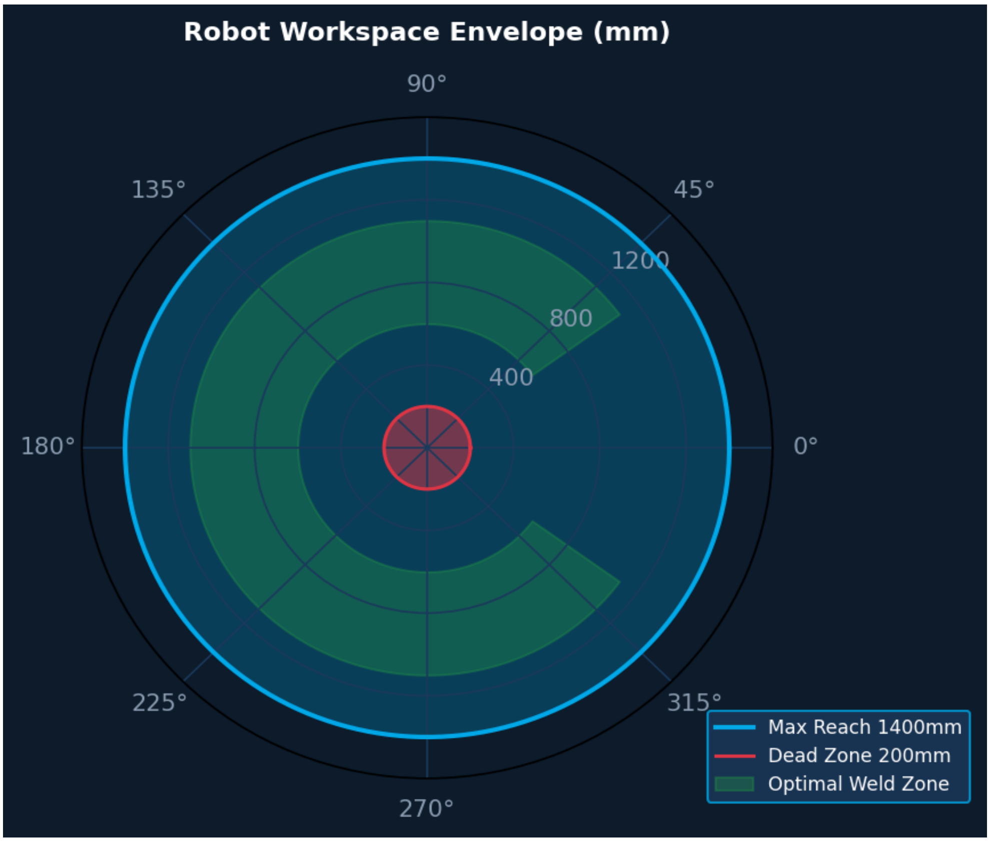

Figure 7 – Robot Workspace Envelope (polar view, mm)

09 Control Algorithms

PID, motion control, path planning and calibration

9.1 PID Control — Joint-Space

Each joint servo loop implements a cascaded PID structure: an outer position loop (500 Hz) feeding an inner velocity loop (2 kHz), which cascades into the drive-internal current loop (16 kHz). Feed-forward gravity and friction compensation terms are computed from the URDF dynamic model.

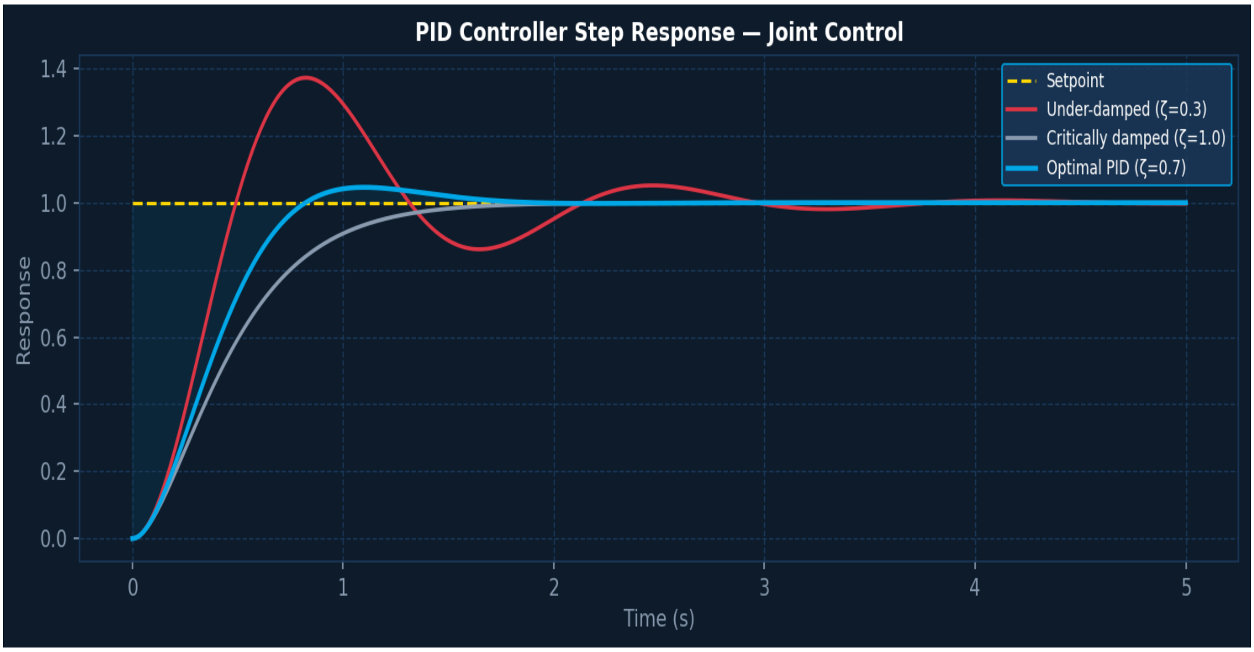

Figure 8 – PID Step Response for Joint Control

9.2 Motion Control

Trajectory execution uses quintic polynomial interpolation between waypoints, with S-curve velocity profiling to limit jerk. Blend radii are automatically computed to maintain TCP speed within ±2% of the commanded value during continuous weld segments.

9.3 Path Planning

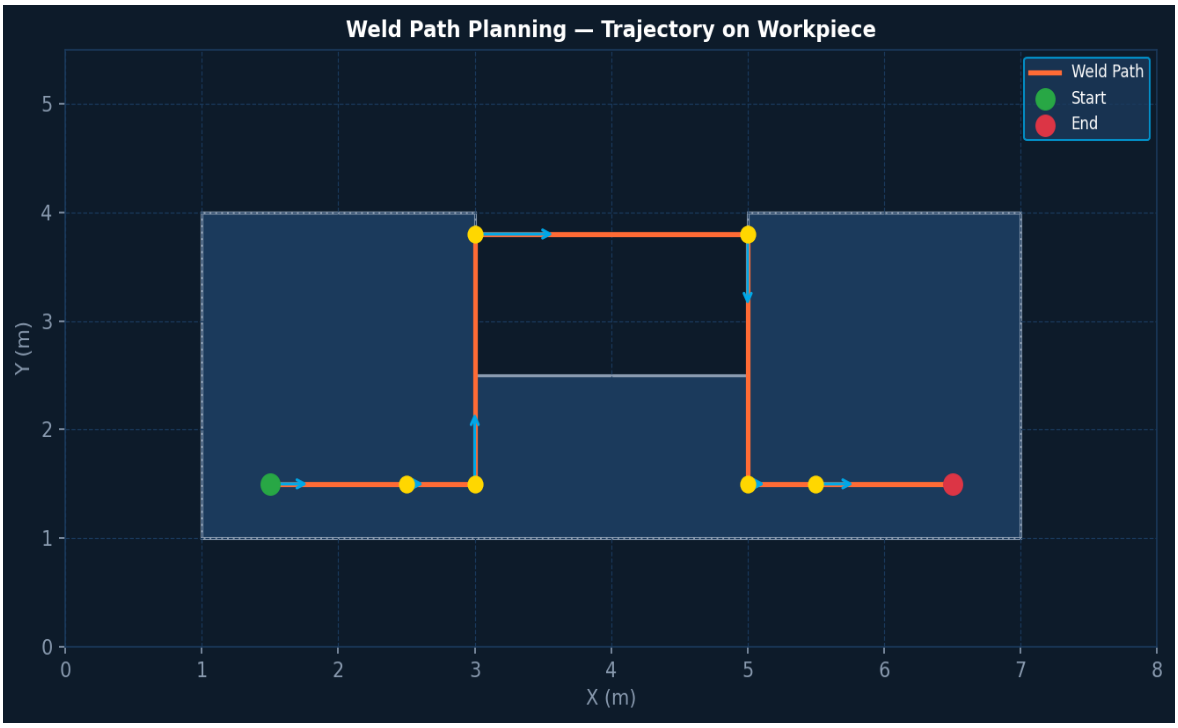

Figure 9 – Weld Path Trajectory on Representative Workpiece

Offline path generation uses a CAD-to-path pipeline: the weld seam is extracted from the STEP/IGES model, discretised into waypoints at ≤ 2 mm intervals, and post-processed with a OMPL-based RRT* planner inside MoveIt 2 to resolve joint-limit and collision constraints.

9.4 Calibration

Kinematic calibration uses a Leica AT960 laser tracker with 20 measurement poses to identify all 24 DH parameter errors. Payload identification uses joint torque measurements during dynamic excitation (IDIM-LS method). Calibration is mandated after any mechanical intervention.

Calibration Schedule	
Full Kinematic Cal.	After mechanical work or collision; annually otherwise
Payload ID	After TCP tooling change
Encoder Reference	At power-on (absolute encoders — no homing required)
TCP Calibration	4-point method; residual < 0.05 mm

10 CAD & Mechanical Design

Design files, tolerances and assembly notes

CAD File Structure

All mechanical design is authored in SolidWorks 2024 and stored under version control (PDM Standard). Native SLDPRT / SLDASM files are the master; STEP 214 exports are generated for downstream simulation and manufacturing.

CAD Reference	
CAD Software	SolidWorks 2024 SP2
PDM System	SolidWorks PDM Standard — "WeldingRobot_WR6A" vault
Export Formats	STEP AP214, IGES, STL (0.01 mm chord deviation)
Drawing Standard	ISO 128 / ISO 2768-m for general tolerances
GD&T;	ISO 1101 — critical bearing bores: ■ H6/j5 fit
FEA	ANSYS 2024 R1 — static + modal, Al7075-T6 material lib
Mass Properties	Maintained in SolidWorks — auto-updated on save

11 Electronics

PCB design, wiring harness and cabinet layout

Electronics Design Overview

Custom PCBs are designed in Altium Designer 24. The main boards are: (1) a breakout board for the 6-axis F/T sensor (SPI → EtherCAT bridge), (2) an I/O expansion board for arc process signals, and (3) a wiring harness interface panel inside the base housing. All PCBs comply with IPC-A-610 Class 3.

Electronics Reference	
Schematic / PCB	Altium Designer 24 — project "WR6A_Electronics"
PCB Class	IPC-A-610 Class 3 — industrial/harsh environment
Connector System	Harting Han 10A (power), Souriau 8STA (signal)
Cable Spec	Igus chainflex CF9 — 10M flex cycles, drag chain rated
EMC Screening	Shielded cables, 360° backshell termination, ferrite cores
Cabinet Cooling	Rittal TopTherm 350 W — thermostat-controlled fan
Cabinet Standard	EN 61439-1 — verified panel builder declaration

13 Manufacturing

Production processes, inspection and quality

Manufacturing Processes

Structural components are 5-axis CNC-machined from billet aluminium (7075-T6) and cast iron. Hard-anodising (25 µm, MIL-A-8625 Type III) is applied to all exposed aluminium surfaces. Gearbox housings are precision-bored post-assembly to ensure bearing seat concentricity < 5 µm.

Quality & Inspection

Dimensional Insp.	Zeiss Contura CMM — ISO 10360-2 — 100% of critical dims
Surface Finish	Ra ≤ 0.8 µm on bearing seats; Ra ≤ 3.2 µm general
Torque Verification	Calibrated torque wrench — all M8+ fasteners logged
Factory Acceptance	8-hour run-in cycle; positional accuracy verified with tracker
Paint / Finish	RAL 7035 powder coat + Loctite 7649 primer on steel
Traceability	Serial numbers laser-engraved; full material certs retained
Standards	ISO 9283 (performance), ISO 10218-1 (safety)

16 Metadata & Revision History

Document control and change log

Document Metadata

Document Number	WR-TECH-2025-001
Title	Welding Robot WR-6A — Technical Documentation
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Approved By	Engineering Director
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Format	PDF/A-1b — archival quality

Rev	Date	Author	Description
0.1	Jan 2025	J. Smith	Draft skeleton — structure agreed
0.5	Mar 2025	A. Kumar	Mechanical, motors and drivers sections added
0.8	Apr 2025	T. Müller	Control algorithms, kinematics, power
0.9	May 2025	J. Smith	Comms architecture, sensors, CAD, electronics
1.0	Jun 2025	All	Final review — approved for release